Priscila Santiesteban

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Academic Interests: As both an educator and a lifelong learner, I work to refine my teaching through classroom experience, educational research, reflection, and student feedback. My teaching emphasizes active engagement, timely feedback, and supporting students as individuals. I am excited to teach across the undergraduate computer science curriculum, with particular interests in introductory programming/computing, data structures and algorithms, and software engineering (including agentic AI for software development).

My research spans the human dimensions of software engineering. I investigate the cognitive (how students and developers think), interactive (how programmers interact with peers and tools), and environmental (how programmers are influenced by their surroundings and behaviors) factors that affect programming outcomes and experiences through rigorous empirical studies, with the goal of improving software engineering practice and informing computer science pedagogy.

Education

Ph.D. in Computer Science and Engineering **Expected May 2027**
University of Michigan Ann Arbor, MI
Advisor: Westley Weimer, Committee Members: Madeline Endres, Amir Kamil, Ioulia Kovelman

M.S. in Computer Science and Engineering **May 2025**
University of Michigan Ann Arbor, MI

B.A. in Computer Science and Physics (Mathematics Minor) **May 2021**
Coe College (a small urban liberal arts college with an enrollment of 1,200 students) Cedar Rapids, IA

Teaching Experience

Instructor on Record **Fall 2025–Winter 2026**
Software Engineering, Upper-Level Undergraduate University of Michigan

Offerings	**Winter 2026**	244 students	Evaluation Avg.: 4.85/5.00, University-Wide: 4.77/5.00
	Fall 2025	244 students	Evaluation Avg.: 4.70/5.00, University-Wide: 4.77/5.00

Format: Taught two sections. Hybrid format (including in-person and asynchronous remote students).

Contributions:

- Designed and delivered lectures in undergraduate software engineering, including developing new instructional materials and in-class activities.
- Created reading-comprehension quizzes to encourage engagement with assigned readings.
- Developed grading rubrics for remote exams to assess authentic student understanding in the presence of generative AI.
- Revised programming assignments to replace obsolete tools and incorporate responsible generative AI use.
- Produced instructional videos explaining course setup, organization, and expectations.
- Refined hybrid participation activities using student feedback to support equitable participation by remote and in-person students.
- Managed a teaching team of four teaching assistants, four graders, and one instructional staff member.
- Revised the course exams in multiple formats (take-home, remote, and in-person) for assessment in the era of generative AI.

Graduate Student Instructor (Teaching Assistant) **Fall 2024**
Software Engineering, Upper-Level Undergraduate University of Michigan

Contributions

- Led discussion sections of up to 14 students.
- Delivered a guest lecture to 290 students.
- Held office hours and supported grading and assessment.

Course Staff **Winter 2025**
Advanced Programming Languages, Graduate-Level University of Michigan

Contributions

- Developed guided lecture notes for a graduate programming languages course with 17 students. This strategy created organizational scaffolding that students identified as helpful for maintaining engagement during lectures.

Undergraduate Teaching Assisstant **Spring 2019**
Introduction to Computer Science I, Required Introductory-Level Coe College

Contributions

- Led weekly laboratory sessions for an undergraduate computer science course of about 30 students, providing one-on-one guidance to students as they worked through coding assignments.

Awards and Honors

Outstanding Graduate Student Instructor Award **Winter 2026**
University of Michigan, Computer Science and Engineering
Awarded for excellence in teaching undergraduates during my appointment as instructor of record, based on student and faculty evaluations.

NSF Graduate Research Fellowship (GRFP) **2023-2026**
National Science Foundation $147,000 Award

Rackham Merit Fellowship **2021-2023**
University of Michigan $91,200 Award
Awarded to students who have outstanding academic qualifications.

NSF Research Experiences for Undergraduates (REU) **Summer 2019, Summer 2020**
Cornell University (2019) and Carnegie Mellon University (2020)

Funding and Grants

Grant Proposal Contributor (Internal University Grant) **2022**
"Understanding the Relationship Between Spatial Reasoning and Programming Using TMS and MRI" Awarded: $17,970
Contributed to the writing, editing, and refinement of a internal grant proposal supporting interdisciplinary research. This project supported multiple work-study students and involved two undergraduate researchers.

Peer-Reviewed Publications

Eight peer-reviewed publications: Four conference papers (ICSE, ESEC/FSE) including a distinguished paper award at ICSE, three journal manuscripts (TSE), and one peer-reviewed workshop paper. Four papers with undergraduate advisees, including two with student first authors. † Indicates an undergraduate mentee.

Conference Proceedings (4)

ICSE '27 **P. Santiesteban**, H. Ahmad, M. Endres, and W. Weimer. ""Hacking the Brain": Improving Coding Outcomes with Neurostimulation" In *ACM/IEEE International Conference on Software Engineering* (Directly accepted in the first round with a 5.0% acceptance rate out of 2,375 submissions; projected max final track acceptance rate 17.6%), 2027.
Accepted for Publication.

ICSE-SEET '26 **P. Santiesteban**, E. Shedden†, M. Endres, and W. Weimer. "An Empirical Study of Anonymous, Unmoderated Online Peer-to-Peer Programming Tutoring Conversations." In *ACM/IEEE Software Engineering Education and Training Track at the International Conference on Software Engineering* (30% acceptance rate), 2026.
https://doi.org/10.1145/3786580.3786958

ICSE '24 H. Ahmad, M. Endres, K. Newman†, **P. Santiesteban**, E. Shedden, and W. Weimer. "Causal Relationships and Programming Outcomes: A Transcranial Magnetic Stimulation Experiment." In *ACM/IEEE International Conference on Software Engineering* (15% acceptance rate), 2026.
ACM SIGSOFT Distinguished Paper Award (10% of accepted papers).
https://doi.org/10.1145/3597503.3639096

ESEC/FSE '21 C. Wong, **P. Santiesteban**, C. Kästner, and C. Le Goues. "VarFix: balancing edit expressiveness and search effectiveness in automated program repair." In *ACM European Software Engineering Conference and Symposium on the Foundations of Software Engineering* (24.49% acceptance rate), 2021.
https://doi.org/10.1145/3468264.3468600

Journal Manuscripts (3)

TSE '26 A. Wei†, **P. Santiesteban**, and W. Weimer. "Sweat the Details: Exercise Breaks and AI Assistance in Code Summarization." In *IEEE Transactions of Software Engineering*, 2026.
https://doi.org/10.1109/TSE.2026.3716079

TSE '24 D. Hu†, **P. Santiesteban**, M. Endres, and W. Weimer. "Towards a Cognitive Model of Dynamic Debugging: Does Identifier Construction Matter?" In *volume 50 issue 11 of IEEE Transactions of Software Engineering*, 2024.
https://doi.org/10.1109/TSE.2024.3465222

TSE '23 **P. Santiesteban**, Y. Huang, W. Weimer, and H. Ahmad. "CirFix: Automated Hardware Repair and its Real-World Applications" In *volume 49 issue 7 of IEEE Transactions of Software Engineering*, 2023.
https://doi.org/10.1109/TSE.2023.3269899

Refereed Workshop Proceedings

GE@ICSE '22 **P. Santiesteban**, M. Endres, and W. Weimer. "An Analysis of Sex Differences in Computing Teaching Evaluations" In the *ACM/IEEE Workshop on Gender Equality, Diversity, and Inclusion in Software Engineering*, 2022.
https://doi.org/10.1145/3524501.3527604

Student Mentoring

Amy Wei **Winter 2025 - Winter 2026**
Undergraduate Research University of Michigan
Advised and mentored on a research project investigating the impact of physical exercise breaks and AI assistance on code summarization performance. A full journal paper where Amy is first author is accepted at TSE 2026.

Emma Shedden **Winter 2023 - Summer 2024**
Undergraduate Research University of Michigan
Assisted on a research project developing a mathematical model of anonymous online peer-to-peer tutoring interactions. A full conference paper where Emma is a co-author is accepted at ICSE-SEET 2026.

Danniell Hu **Winter 2023 - Fall 2023**
Undergraduate Research University of Michigan
Advised on a research project investigating the neural correlates of dynamic debugging using medical imaging. A full journal paper where Danni is first author is accepted at TSE 2024.

Kaia Newman **Spring 2023 - Summer 2023**
Undergraduate Research (Secondary Advisor) University of Michigan
Assisted on a research project investigating the causal relationship between code comprehension and spatial ability. A full conference paper where Kaia is a co-author is accepted at ICSE 2024 (Distinguished Paper Award).

Joel Estrada and Syed Khan **Winter 2026**
Senior Independent Study University of Michigan
Supervised both undergraduate students through completion of a senior independent study project focused on applying machine learning to finance. Both students successfully wrote the thesis, completed the study, and graduated.

Service

Peer Review

Journal Manuscript Reviewer **Fall 2025**
Publishing Research Quarterly (Springer Nature)
Reviewed a submitted manuscript in publishing and literary studies.

University Service

Research Workshop Mentor **Fall 2025**
University of Michigan
Mentored approximately 50 incoming undergraduate students through a research workshop.

Judge, Undergraduate Research Symposium **Winter 2026**
University of Michigan
Evaluated 25 undergraduate student research presentations.

Graduate Recruitment Panelist **Winter 2022, Winter 2023**
University of Michigan
Participated in two panel discussions to an audience of 20 admitted Computer Science and Engineering Ph.D. students each, answering questions about research, graduate life, and the doctoral program.

Community Resources

Maintainer, CS Graduate School Application Guide **Fall 2025 - Present**
Publicly Online
Co-developed, deployed, and maintain a publicly–available educational resource to support prospective student applicants to computer science graduate programs. The guide synthesizes faculty perspectives, graduate student experiences, application advice, and curated resources covering Master's and Ph.D. admissions. Since its launch in late 2025, the guide has served users across five countries. https://csguides.github.io/grad-school-application-guide/

Teaching Development

Workshops (Winter 2024)	Completed five professional development workshops through the Michigan Center for Research on Learning and Teaching, including: "Are They Getting It?: Low-Stakes Ways to Gauge Student Learning", "Making Groupwork Work", "Strategies for Leading Office Hours in Engineering Courses", "Introduction to Neurodivergence: Strategies for Engineering Instructors", and "The Importance of Embracing Failure in Learning".

Miscellaneous

Languages	Native English, Native Spanish
Citizenship	United States

References

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